01

## MASTER VOCABULARY: WHAT EXACTLY ARE WE STUDYING?

**CONTEXT + EXACT CORE**

Structure → process → surface: lithology, bedding, joints, faults and tectonic setting create a template; weathering, rivers, glaciers, waves and human use modify it.

**MECHANISM / ARGUMENT**

A rock aggregate is therefore not "a mineral", and a landform cannot be inferred from a rock name alone.

**CONSEQUENCE / CONTRAST**

Structure → process → surface: lithology, bedding, joints, faults and tectonic setting create a template; weathering, rivers, glaciers, waves and human use modify it.

**UPSC TRAP / ANSWER USE**

A rock aggregate is therefore not "a mineral", and a landform cannot be inferred from a rock name alone.

**ANSWER LINE:** **Geological structure is the age, composition and architecture of rocks and tectonic features beneath a region, while physiography is their broad surface expression and geomorphology explains the processes that reshape it.**

02

## CRUST AND LITHOSPHERE: RELATED, NOT IDENTICAL

**CONTEXT + EXACT CORE**

Moho means the crust–mantle seismic boundary; its wave evidence belongs to Topic 01 and is only cross-linked here.

**MECHANISM / ARGUMENT**

Asthenosphere means the weaker, slowly deforming upper-mantle zone beneath the lithosphere; plates move as coherent lithospheric units over it.

**CONSEQUENCE / CONTRAST**

Crust is compositional; lithosphere is mechanical and includes the crust plus rigid uppermost mantle.

**UPSC TRAP / ANSWER USE**

They are useful mnemonics, not complete modern compositional layers.

**ANSWER LINE:** **The crust is Earth's outer compositional shell, whereas the lithosphere is the stronger mechanical layer comprising the crust plus the rigid uppermost mantle.**

03

## THE COMPLETE ROCK CYCLE: A NETWORK, NOT A CONVEYOR BELT

**CONTEXT + EXACT CORE**

Lithification means turning loose sediment into rock through compaction and cementation.

**MECHANISM / ARGUMENT**

Lithification means turning loose sediment into rock through compaction and cementation.

**CONSEQUENCE / CONTRAST**

"Igneous → sedimentary → metamorphic" is one teaching path, not the only path.

**UPSC TRAP / ANSWER USE**

Metamorphism normally occurs without complete melting; if melting occurs, the material re-enters the igneous path.

**ANSWER LINE:** **The rock cycle is an open network in which melting, crystallisation, weathering, lithification, metamorphism and uplift can connect any rock family through multiple pathways rather than one fixed sequence.**

04

## IGNEOUS ROCKS: COOLING HISTORY, COMPOSITION AND TEXTURE

**CONTEXT + EXACT CORE**

"Primary rock" is an older genetic phrase for igneous rock because crystallisation creates the original mineral framework.

**MECHANISM / ARGUMENT**

Porphyritic means large crystals embedded in a finer groundmass, showing two-stage cooling.

**CONSEQUENCE / CONTRAST**

It does not mean every igneous exposure is oldest or economically primary.

**UPSC TRAP / ANSWER USE**

India anchor: Deccan basalt is extrusive flood basalt; Peninsular granites and gneisses have different histories and should not be merged with the Deccan province.

**ANSWER LINE:** **"Primary rock" is an older genetic phrase for igneous rock because crystallisation creates the original mineral framework.**

05

## SEDIMENTARY ROCKS: SEDIMENT PATHWAY AND FIELD CLUES

**CONTEXT + EXACT CORE**

Fossils, ripple marks, mud cracks and cross-bedding can reveal environment, current direction, drying or younging.

**MECHANISM / ARGUMENT**

Fossils, ripple marks, mud cracks and cross-bedding can reveal environment, current direction, drying or younging.

**CONSEQUENCE / CONTRAST**

Limestone has several origins; do not call every limestone exclusively organic or chemical.

**UPSC TRAP / ANSWER USE**

Loess is wind-deposited silt and moraine is glacial debris.

**ANSWER LINE:** **Conglomerate has rounded gravel clasts; sandstone is sand-sized; shale is fine mud-rich rock.**

06

## METAMORPHIC ROCKS: PROTOLITH, GRADE AND FABRIC

**CONTEXT + EXACT CORE**

Metamorphic grade means intensity of temperature–pressure conditions.

**MECHANISM / ARGUMENT**

Metamorphic grade means intensity of temperature–pressure conditions.

**CONSEQUENCE / CONTRAST**

Graphite may form from carbon-rich material, but "coal → graphite" is not a universal one-step rule.

**UPSC TRAP / ANSWER USE**

Agents are heat, confining pressure, differential stress—unequal directed pressure—and chemically active fluids.

**ANSWER LINE:** **Exam-safe paths: shale/mudstone → slate → phyllite → schist → gneiss is a common increasing-grade teaching sequence, not a universal compulsory path.**

07

## ROCK PROPERTIES AND APPLIED GEOGRAPHY

**CONTEXT + EXACT CORE**

Rock name alone never determines land use: porosity, permeability, fractures, weathering, structure and site conditions jointly control groundwater, soils, foundations, dams and slope behaviour.

**MECHANISM / ARGUMENT**

Crystalline igneous and metamorphic rocks commonly have low primary pore space but can store groundwater in weathered zones, joints and fractures.

**CONSEQUENCE / CONTRAST**

Basalt aquifers are heterogeneous: vesicular tops and flow contacts may transmit water; compact interiors may not.

**UPSC TRAP / ANSWER USE**

A dam, tunnel or foundation requires site investigation; "granite is strong" or "limestone is weak" is not an engineering conclusion.

**ANSWER LINE:** **Rock name alone never determines land use: porosity, permeability, fractures, weathering, structure and site conditions jointly control groundwater, soils, foundations, dams and slope behaviour.**

08

## CONTINENTAL DRIFT: WEGENER'S EVIDENCE AND LIMITATION

**CONTEXT + EXACT CORE**

Continental drift became persuasive because shelf-edge fit, matching rock belts, fossils, palaeoclimate and palaeomagnetism independently point to former continental connections, even though Wegener lacked an adequate mechanism.

**MECHANISM / ARGUMENT**

It must later be distinguished from India's Gondwana Supergroup of rocks.

**CONSEQUENCE / CONTRAST**

His evidence was powerful, but proposed driving forces were inadequate.

**UPSC TRAP / ANSWER USE**

Gondwana here means a southern palaeocontinent.

**ANSWER LINE:** **Continental drift became persuasive because shelf-edge fit, matching rock belts, fossils, palaeoclimate and palaeomagnetism independently point to former continental connections, even though Wegener lacked an adequate mechanism.**

09

## SEA-FLOOR SPREADING: OCEANIC EVIDENCE AND MASS BALANCE

**CONTEXT + EXACT CORE**

Sea-floor spreading creates young oceanic crust at mid-ocean ridges, records symmetrical magnetic stripes and balances creation by consuming older lithosphere at trenches, so the Earth does not expand.

**MECHANISM / ARGUMENT**

Ocean floor is geologically young relative to continental shields because it is recycled.

**CONSEQUENCE / CONTRAST**

The Earth does not expand; ridge creation is balanced by trench destruction.

**UPSC TRAP / ANSWER USE**

Magnetic reversals create matching stripe patterns on both ridge flanks.

**ANSWER LINE:** **Sea-floor spreading creates young oceanic crust at mid-ocean ridges, records symmetrical magnetic stripes and balances creation by consuming older lithosphere at trenches, so the Earth does not expand.**

10

## PLATE TECTONICS: BOUNDARY MATRIX AND LANDFORM DIAGNOSIS

**CONTEXT + EXACT CORE**

Plate-interior activity occurs at inherited faults, rifts, hot spots and intraplate stress concentrations; boundaries explain global clustering, not every event.

**MECHANISM / ARGUMENT**

Plate-interior activity occurs at inherited faults, rifts, hot spots and intraplate stress concentrations; boundaries explain global clustering, not every event.

**CONSEQUENCE / CONTRAST**

Andes and Himalaya are both mountains but arise from different convergence types.

**UPSC TRAP / ANSWER USE**

Plate-interior activity occurs at inherited faults, rifts, hot spots and intraplate stress concentrations; boundaries explain global clustering, not every event.

**ANSWER LINE:** **Plate boundaries are diagnosed by motion and products: divergence creates crust, subduction destroys oceanic crust, continent collision thickens crust, and transform motion conserves crust while generating shallow earthquakes.**

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## OCEAN-BASIN EVOLUTION: WILSON-CYCLE ANALOGY

**CONTEXT + EXACT CORE**

Passive margins can accumulate thick sediment wedges and host offshore petroleum systems.

**MECHANISM / ARGUMENT**

Passive margins can accumulate thick sediment wedges and host offshore petroleum systems.

**CONSEQUENCE / CONTRAST**

A basin can stall, reorganise or avoid the ideal sequence; the examples are analogies, not timestamps.

**UPSC TRAP / ANSWER USE**

The Wilson cycle is an analogy for rifting, ocean opening, subduction, closure and suturing, not a fixed destiny that every ocean basin must complete.

**ANSWER LINE:** **The Wilson cycle is an analogy for rifting, ocean opening, subduction, closure and suturing, not a fixed destiny that every ocean basin must complete.**

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## ISOSTASY AND OROGENY

**CONTEXT + EXACT CORE**

Isostasy explains vertical crustal adjustment, while orogeny explains mountain building through shortening, thickening and uplift; erosion and unloading can then trigger further isostatic response.

**MECHANISM / ARGUMENT**

Orogeny means mountain building through convergence, folding, thrusting, metamorphism, crustal thickening and uplift.

**CONSEQUENCE / CONTRAST**

Erosion removes mass, exposes deeper rocks and can produce isostatic response; high relief is therefore dynamic, not a one-time construction.

**UPSC TRAP / ANSWER USE**

Isostasy means buoyant vertical equilibrium between crust/lithosphere and denser material beneath.

**ANSWER LINE:** **Isostasy explains vertical crustal adjustment, while orogeny explains mountain building through shortening, thickening and uplift; erosion and unloading can then trigger further isostatic response.**

13

## WORLD-TO-INDIA TRANSITION: STRUCTURE EXPRESSED AS RELIEF

**CONTEXT + EXACT CORE**

Structure controls broad relief and resource opportunity, while climate, weathering, rivers and human use modify it.

**MECHANISM / ARGUMENT**

At the highest school-geography scale, India's six broad physiographic divisions are the Himalaya/Northern Mountains, Northern Plain, Peninsular Plateau, Indian Desert, Coastal Plains and Islands.

**CONSEQUENCE / CONTRAST**

They are used here only as structural expressions, not as substitutes for the later detailed physiography owners.

**UPSC TRAP / ANSWER USE**

Structure controls broad relief and resource opportunity, while climate, weathering, rivers and human use modify it.

**ANSWER LINE:** **Structure controls broad relief and resource opportunity, while climate, weathering, rivers and human use modify it.**

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## SCOPE: GEOLOGICAL STRUCTURE VERSUS PHYSIOGRAPHY AND GEOMORPHOLOGY

**CONTEXT + EXACT CORE**

Geological structure means the age, lithology, stratigraphic order, folds, faults, joints, basins, cratons and plate-tectonic relations of rocks beneath a region.

**MECHANISM / ARGUMENT**

This package supplies their geological mechanism and clearly identifies those cross-links.

**CONSEQUENCE / CONTRAST**

Exam method: identify the hidden structure, name a region, explain the mechanism, state a surface/resource/hazard consequence, then qualify the relationship.

**UPSC TRAP / ANSWER USE**

Therefore geology explains pattern and constraint, not every present landform by itself.

**ANSWER LINE:** **India's geological skeleton consists of an ancient Peninsular shield, an active Himalayan fold-thrust belt and a sediment-filled Indo-Ganga-Brahmaputra foreland basin linked by collision, flexure, erosion and deposition.**

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## GEOLOGICAL TIME, EVIDENCE AND DATING CAUTIONS

**CONTEXT + EXACT CORE**

Mains use: Use an evidence ladder: field relation - dating/correlation - geophysical image - tectonic interpretation - qualification.

**MECHANISM / ARGUMENT**

Relative dating uses superposition, cross-cutting relations, unconformities, fossils and correlation.

**CONSEQUENCE / CONTRAST**

Fossils are especially useful in sedimentary correlations, but many ancient Indian crystalline and Proterozoic successions are unfossiliferous or sparsely fossiliferous.

**UPSC TRAP / ANSWER USE**

Stratigraphy is not merely a rock-name list.

**ANSWER LINE:** **A safe geological chronology combines superposition, cross-cutting relations, unconformities, fossils, radiometric event ages, palaeomagnetism and geophysics; relative order is often safer than memorised numerical ages.**

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## INDIAN PLATE: GONDWANA ORIGIN, BREAK-UP, DRIFT AND COLLISION

**CONTEXT + EXACT CORE**

The Indian plate journey links Gondwana rifting, northward motion, Tethyan ocean closure and prolonged India-Eurasia collision to Himalayan uplift, foreland subsidence and continuing seismicity.

**MECHANISM / ARGUMENT**

Continuing convergence is expressed through active thrusting, earthquakes, uplift, erosion and sediment transfer into the foreland and offshore basins.

**CONSEQUENCE / CONTRAST**

Collision was not a single instant across the whole margin.

**UPSC TRAP / ANSWER USE**

India is both an ancient continental fragment and part of an active plate system: this explains the contrast between old shield rocks and young Himalayan tectonic architecture.

**ANSWER LINE:** **The Indian plate journey links Gondwana rifting, northward motion, Tethyan ocean closure and prolonged India-Eurasia collision to Himalayan uplift, foreland subsidence and continuing seismicity.**

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## CRATONS, SHIELDS AND THE STABILISED PENINSULAR BLOCK

**CONTEXT + EXACT CORE**

Peninsular India is a mosaic of ancient cratonic nuclei and intervening mobile belts, later cut by faults, rifts and dykes and locally covered by younger sediments and basalt.

**MECHANISM / ARGUMENT**

Peninsular India is a mosaic of ancient cratonic nuclei and intervening mobile belts, later cut by faults, rifts and dykes and locally covered by younger sediments and basalt.

**CONSEQUENCE / CONTRAST**

Metallic-mineral provinces are strongly associated with ancient shield geology, but ore localisation requires specific magmatic, hydrothermal, sedimentary or metamorphic processes.

**UPSC TRAP / ANSWER USE**

It does not mean flat, homogeneous, immobile or earthquake-free.

**ANSWER LINE:** **Peninsular India is a mosaic of cratons and mobile belts later cut by faults, rifts, dykes and basins and partly covered by younger sediments and lava, so stable never means uniform, flat or earthquake-free.**

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## ARCHAEAN AND DHARWAR SYSTEMS: DISTRIBUTION AND TERMINOLOGY CONTROL

**CONTEXT + EXACT CORE**

Mains use: In a classification answer, give lithology, structure, distribution, significance and a terminology caution.

**MECHANISM / ARGUMENT**

Mains use: In a classification answer, give lithology, structure, distribution, significance and a terminology caution.

**CONSEQUENCE / CONTRAST**

Do not repeat old claims that all Archaean rocks are entirely unfossiliferous or that one textbook's age span is definitive.

**UPSC TRAP / ANSWER USE**

Named examples such as Kolar or Singhbhum are location anchors, not universal rules for every Dharwar rock.

**ANSWER LINE:** **Archaean granitoid-gneiss basement and Dharwar-type supracrustal belts form distinct parts of the ancient shield, and their ore associations reflect specific magmatic, sedimentary, metamorphic and hydrothermal histories rather than age alone.**

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## PROTEROZOIC PURANA BASINS: CUDDAPAH AND VINDHYAN

**CONTEXT + EXACT CORE**

Resource links must be bounded: a whole system is not uniformly mineralised, and Panna diamonds do not make every Vindhyan sandstone diamond-bearing.

**MECHANISM / ARGUMENT**

Cuddapah and Vindhyan are major Proterozoic sedimentary basin successions resting unconformably on older basement.

**CONSEQUENCE / CONTRAST**

Resource links must be bounded: a whole system is not uniformly mineralised, and Panna diamonds do not make every Vindhyan sandstone diamond-bearing.

**UPSC TRAP / ANSWER USE**

Older textbooks group them under Purana, but their detailed ages and correlations are convention-sensitive.

**ANSWER LINE:** **Purana is an older Indian label for Proterozoic basin successions such as Cuddapah and Vindhyan, which rest unconformably on older basement and record long-lived sedimentation rather than one uniform mineral province.**

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## GONDWANA SUPERGROUP: RIFT BASINS, COAL AND CONTINENTAL SEDIMENTATION

**CONTEXT + EXACT CORE**

Lower parts of the succession include glacially influenced deposits in some basins, followed by coal-bearing sandstone-shale cycles and younger continental strata.

**MECHANISM / ARGUMENT**

Lower parts of the succession include glacially influenced deposits in some basins, followed by coal-bearing sandstone-shale cycles and younger continental strata.

**CONSEQUENCE / CONTRAST**

Coal presence is not automatic in every Gondwana unit or every rift.

**UPSC TRAP / ANSWER USE**

Gondwana coalfields are not simply surface strips along modern rivers.

**ANSWER LINE:** **The Gondwana Supergroup is a fault-basin continental sedimentary succession with important coal measures; it must not be confused with Gondwana, the southern palaeocontinent.**

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## MARINE SEDIMENTARY BASINS, COASTS AND OFFSHORE PETROLEUM LOGIC

**CONTEXT + EXACT CORE**

Mains use: Explain offshore concentration through basin evolution, not a memorised field list.

**MECHANISM / ARGUMENT**

Mains use: Explain offshore concentration through basin evolution, not a memorised field list.

**CONSEQUENCE / CONTRAST**

Do not claim every named basin is equally productive or that all offshore oil is young.

**UPSC TRAP / ANSWER USE**

Thick sediment alone is necessary in many settings but never sufficient.

**ANSWER LINE:** **Offshore occurrence is therefore geological, not simply because an area is under the sea.**

22

## DECCAN TRAP: FLOOD-BASALT PROVINCE AND BOUNDED SURFACE LINKAGES

**CONTEXT + EXACT CORE**

The Deccan Trap is a vast flood-basalt province built by repeated, mainly fissure-fed lava flows around the end-Cretaceous to earliest Cenozoic transition.

**MECHANISM / ARGUMENT**

The Deccan Trap is a vast flood-basalt province built by repeated, mainly fissure-fed lava flows around the end-Cretaceous to earliest Cenozoic transition.

**CONSEQUENCE / CONTRAST**

Deccan Trap is a volcanic province, not a soil type.

**UPSC TRAP / ANSWER USE**

This architecture governs groundwater: compact basalt may transmit little water, while fractures, vesicular zones, weathered contacts and interflow horizons can form aquifers.

**ANSWER LINE:** **The Deccan Trap is a Late Cretaceous flood-basalt province of stacked fissure-fed flows whose contacts, fractures, vesicular zones and weathering create step relief, black-soil parent material and highly variable aquifers.**

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## HIMALAYA: FROM GEOSYNCLINAL LANGUAGE TO MODERN PLATE TECTONICS

**CONTEXT + EXACT CORE**

Modern plate tectonics explains the driving mechanism through subduction of oceanic domains, collision, shortening, thrusting, metamorphism and uplift.

**MECHANISM / ARGUMENT**

Modern plate tectonics explains the driving mechanism through subduction of oceanic domains, collision, shortening, thrusting, metamorphism and uplift.

**CONSEQUENCE / CONTRAST**

Mains use: Contrast the historical geosynclinal description with the modern plate-tectonic mechanism; then draw a labelled cross-section.

**UPSC TRAP / ANSWER USE**

They are not the oldest Himalayan rocks.

**ANSWER LINE:** **The Himalaya is a continent-collision fold-thrust orogen containing sedimentary, metamorphic and crystalline belts; unlike the Andes, it is not primarily a volcanic subduction arc.**

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## INDO-GANGA-BRAHMAPUTRA FORELAND BASIN AND ALLUVIAL FILL

**CONTEXT + EXACT CORE**

Correct: It is a sediment-filled foreland basin.

**MECHANISM / ARGUMENT**

Correct: It is a sediment-filled foreland basin.

**CONSEQUENCE / CONTRAST**

Thick unconsolidated sediment explains level relief, fertile alluvial soils, major aquifers and dense settlement, but also flood, erosion, subsidence and liquefaction sensitivity.

**UPSC TRAP / ANSWER USE**

The plain is not one uniform layer or one age.

**ANSWER LINE:** **The Indo-Ganga-Brahmaputra Plain is not structureless alluvium but a flexural foreland basin whose thick sediment supports fertile soils and aquifers while also influencing floods, subsidence and seismic site response.**

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## COASTS, OFFSHORE BASINS AND ISLAND-ARC VERSUS CORAL-ISLAND DISTINCTION

**CONTEXT + EXACT CORE**

Andaman-Nicobar belongs to an active subduction-related arc-forearc system, whereas Lakshadweep consists of coral atolls on a submarine volcanic-ridge foundation; Indian island geology is therefore not uniform.

**MECHANISM / ARGUMENT**

The emergent islands are coral, unlike the tectonic island-arc setting of Andaman-Nicobar.

**CONSEQUENCE / CONTRAST**

The island chain also includes sedimentary and uplifted components; it is not simply a row of volcanic cones.

**UPSC TRAP / ANSWER USE**

Correct: Lakshadweep is coral; Andaman-Nicobar has an active tectonic arc setting.

**ANSWER LINE:** **Andaman-Nicobar belongs to an active subduction-related arc-forearc system, whereas Lakshadweep consists of coral atolls on a submarine volcanic-ridge foundation; Indian island geology is therefore not uniform.**

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## MAJOR LINEAMENTS, FAULTS AND RIFTS

**CONTEXT + EXACT CORE**

Mains use: Use one map plus one mechanism paragraph; avoid asserting current activity without evidence.

**MECHANISM / ARGUMENT**

Mains use: Use one map plus one mechanism paragraph; avoid asserting current activity without evidence.

**CONSEQUENCE / CONTRAST**

A lineament mapped from imagery is not automatically a proven active fault.

**UPSC TRAP / ANSWER USE**

Schematic national maps must not imply exact fault traces or legal boundary precision.

**ANSWER LINE:** **A rift is an extensional crustal zone, a graben is a down-dropped fault block, and a lineament is a mapped linear feature that is not automatically an active fault.**

27

## STRUCTURAL CONTROL OF RELIEF, DRAINAGE, GROUNDWATER AND WATERFALLS

**CONTEXT + EXACT CORE**

Structure guides but does not fully fix drainage.

**MECHANISM / ARGUMENT**

Hard-rock groundwater depends on secondary openings.

**CONSEQUENCE / CONTRAST**

Correct: Lithologic contrast, rejuvenation and lava contacts are alternative controls.

**UPSC TRAP / ANSWER USE**

Structure guides but does not fully fix drainage.

**ANSWER LINE:** **Geological structure becomes examinable when it explains why relief, rivers, groundwater and resource patterns are spatially organised.**

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## GEOLOGY, MINERALS, COAL AND PETROLEUM: ASSOCIATION WITHOUT DETERMINISM

**CONTEXT + EXACT CORE**

Petroleum and gas occur in sedimentary basins onshore and offshore where a complete petroleum system developed and survived.

**MECHANISM / ARGUMENT**

India's resource map reflects ancient shield mineralisation, Gondwana coal basins, sedimentary petroleum systems, coastal placers and weathering-derived ores.

**CONSEQUENCE / CONTRAST**

It does not imply that all critical minerals lie in one geological province.

**UPSC TRAP / ANSWER USE**

NCMM is a current policy anchor, not a geological classification.

**ANSWER LINE:** **Shield belts, Gondwana basins and sedimentary margins are associated respectively with many metallic ores, coal and petroleum, but exploration, grade, preservation, technology, economics and environmental limits decide usable resources.**

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## SEISMICITY: HIMALAYA, KACHCHH AND PENINSULAR INTRAPLATE EVENTS

**CONTEXT + EXACT CORE**

India's seismicity reflects active plate-boundary deformation in the Himalaya and north-east, subduction near Andaman-Nicobar, and intraplate reactivation in Kachchh and parts of the Peninsula.

**MECHANISM / ARGUMENT**

India's seismicity reflects active plate-boundary deformation in the Himalaya and north-east, subduction near Andaman-Nicobar, and intraplate reactivation in Kachchh and parts of the Peninsula.

**CONSEQUENCE / CONTRAST**

Mains use: Explain source, path/site effect and vulnerability separately; do not equate tectonic activity with disaster magnitude.

**UPSC TRAP / ANSWER USE**

Kachchh is an intraplate high-hazard region with inherited rift faults and damaging historical earthquakes, including the 2001 Bhuj event.

**ANSWER LINE:** **India's seismicity reflects active Himalayan and Andaman plate boundaries plus reactivation of inherited structures such as Kachchh and selected Peninsular faults; seismic zoning is a design generalisation, not earthquake prediction.**